

The Acadian Forest of New Brunswick in the 21st century: what shifting heat and water balance imply for future stand dynamics and management

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Abstract

Climate change is altering the dynamics of New Brunswick's forests. To mitigate the effects of climate change, it is crucial to account for future uncertainties in climate projections and natural disturbance trajectories when designing forestry practices. This paper presents the projected changes in key climatic drivers for New Brunswick's forests, examines the impacts of climate change on forest stand dynamics, and reviews adaptive silviculture tools for climate change adaptation. By 2071–2100, a projected 4–6 °C increase in mean annual temperature will lead to a 39%–77% rise in growing degree-days and a reduction in summer atmospheric water balance by 48–79 mm across New Brunswick. Foresters should anticipate a doubling of annual area burned, the northward migration of spruce budworm (*Choristoneura fumiferana* (Clemens)), and the introduction of novel insects and diseases. Forest simulation models project a severe decline in boreal tree species abundance, including a 50% decline in balsam fir (*Abies balsamea* (L.) Mill) and black spruce (*Picea mariana* (Mill.) Britton et al.), offset by an increase in temperate species, notably red maple (*Acer rubrum* L.; +200%) and American beech (*Fagus grandifolia* Ehrh; +135%). Forests in the highlands and lowlands ecoregions, with 40%–50% and 15%–30% spruce and balsam fir composition, respectively, are particularly vulnerable. To limit climate change effects on forest values, foresters must prioritize climate resiliency in their management plans. Strategies should ensure the steady provision of forest goods and services under changing climatic conditions. Forest thinning enhances stand productivity and resilience, while shelterwood and two-age harvest balances timber production with diversity. Clearcutting, despite drawbacks, establishes younger, more vigorous forests with higher carbon sequestration potential. Assisted migration offers promise in helping threatened tree species adapt. Climate change will have a significant impact on New Brunswick's forests, leading to changes in key climatic drivers, increased risks of disturbances, and a shift toward more temperate tree species. However, there are existing adaptation strategies available. It is crucial to consider future uncertainties when designing and evaluating forestry practices, as this is essential for mitigating the effects of climate change on forest values and ensuring the continued provision of forest goods and services over time.

Key words: climate change, climate moisture index, composition shift, natural disturbances, forest resilience

1. Introduction

Climate change is altering the structure and function of forest ecosystems globally and the frequency, severity, and extent of natural disturbances (e.g., fire, insect outbreaks, droughts, and windthrow; Allen et al. 2010). To maintain the production of goods and services we derive from forests, modern foresters are tasked with accounting for future climate change impacts on forest stand and landscape dynamics in their management plans (Dale et al. 2001). Key to maintaining this production and ecosystem services (e.g., timber, carbon sequestration, wildlife habitat, etc.) in a changing climate is the maintenance of forest conditions that are resilient to change (also known as resilient forests; Messier et al. 2019). Forest resilience refers to the ability of

a forest ecosystem to withstand and recover from disturbances, such as wildfires, insect outbreaks, extreme weather events, or human interventions, while maintaining its essential functions and structure. It involves the capacity of forests to absorb and adapt to changes (Nagel et al. 2017; Messier et al. 2019). The concept of forest resilience can provide a useful perspective from which to evaluate the suitability of forest management strategies over time, especially under climate change (Bryant et al. 2019). Challenges in applying this concept are (1) establishing what forest stand conditions are resilient to change and (2) developing silvicultural approaches to guide the forest toward these conditions. Most resiliency studies often focus on single disturbances in a particular forest type, whereas forested landscapes are generally

subject to multiple disturbance types (Seidl et al. 2017; Taylor et al. 2020b). Hence broader knowledge of forest resilience is limited. However, resiliency studies can be used to build distinct forest resilience indicators to inform planning models. Resilience indicators are quantitative or qualitative measures used to assess the ability of a forest ecosystem to withstand and recover from disturbances (Bryant et al. 2019). These indicators provide insights into the health, adaptability, and overall robustness of a forest, enabling forest managers and researchers to monitor and evaluate its capacity to maintain ecological functions and services over time. Forest resilience indicators may include measures of biodiversity, species composition, forest structure, regeneration capacity, soil health, carbon storage, and the presence of key ecological processes (Dymond et al. 2015; Bryant et al. 2019). By monitoring these indicators, stakeholders can make informed decisions and implement appropriate management strategies to enhance the long-term sustainability of forest ecosystems.

In the province of New Brunswick, Canada, forests cover 84% of the provincial landmass (Nadeau, et al. 2008) and the forest industry generates 1.7 billion dollars annually (GDP) from pulp and paper and other wood products manufacturing (Forest NB 2018). New Brunswick is part of the Acadian Forest Region (Rowe 1972), a unique ecological transition zone that bridges boreal forests to the north and temperate forests to the south. As a result, many constituent tree species are at their extreme northern and southern geographical range limits. Such transition zones are considered highly sensitive to climate change (Fisichelli et al. 2014; Evans and Brown 2017). The Acadian Forest is composed of approximately 65% conifers and 35% hardwood tree species but dominated by balsam fir (*Abies balsamea* (L.) Mill), black spruce (*Picea mariana* (Mill) Britton et al.), red maple (*Acer rubrum* L.), trembling poplar (*Populus tremuloides* (Michx.)), and balsam poplar (*Populus balsamifera* (L.)).

Addressing regional climate change impacts requires an understanding of how heat and moisture are projected to change over time. Mean temperatures in New Brunswick have increased by over 1.5 °C since the early 20th century (1900–2022; NOAA 2022). These elevated temperatures have caused reductions in water availability by increasing evapotranspiration by 1 mm year⁻¹ over the 1960–2000 period (Fernandes et al. 2007) and have amplified the growing season length by over 20 days (Zhang et al. 2019). These trends are expected to accelerate over the next century (Zhang et al. 2019). Regional forest projections using the stand model PICUS (named after a Roman god of the forests) predict reductions of 25%–60% in the growth of dominant boreal species in the coming decades, which will not be compensated by increasing growth in temperate species (Boulanger et al. 2017; Taylor et al. 2017). This net growth reduction will cause commercial species aboveground biomass to drop by over 25% by the year 2100 across the region under the high emission scenario, RCP 8.5 (Brecka et al. 2020; Taylor et al. 2017). A portion of decline in commercial tree biomass is likely to be attributed to the projected productivity gains in some shrub species (e.g., leatherleaf (*Chamaedaphne calyculata* (L.) Moench) and Labrador tea (*Rhododendron groenlandicum* (Oeder) Kron &

Judd)) in the region (Ward et al. 2019), raising serious wood supply concerns.

A growing body of evidence suggests that future productivity rates and natural disturbance dynamics are shifting away from historical baselines (McLain and Lee 1996; Taylor et al. 2020b). Establishing successful silvicultural strategies that guide forest conditions that are resilient to these changing dynamics requires large-scale collaboration and intensive monitoring systems (Noulèkoun et al. 2021). Various forms of “adaptive silviculture to climate change” are currently being tested on multiple sites across the US (e.g., Lu et al. 2014; Nagel et al. 2017; Knapp et al. 2021) and more extensively in the Canadian boreal forest (e.g., Sang et al. 2019). Several possible adaptive silviculture techniques are gaining interest, including (1) assisted tree species migration, i.e., the human-assisted movement of tree species and genotypes in areas outside of their known geographical range to overcome slow natural dispersal (Dumroese 2013); (2) thinning (pre-commercial and commercial), used to manage composition toward increasingly climate resilient species, impacting density to promote growth and survival of selected trees (Navarro-Cerrillo et al. 2019); and (3) shelterwood and two-age harvesting treatments, aiming respectively at an even and two-aged class structure while using the canopy of mature trees to act as physical buffer protecting the understory against the shifting climate (Wang et al. 2015a).

The purpose of this study is to (1) summarize and report on projected changes in climatic variables relevant to forest management across New Brunswick’s forests; (2) review projected changes in the structure, function, and dynamics of New Brunswick’s forests; and (3) review novel forest resilience indicators and silvicultural interventions that are currently being promoted and tested in climate change resiliency trials, and their applicability to New Brunswick’s forest.

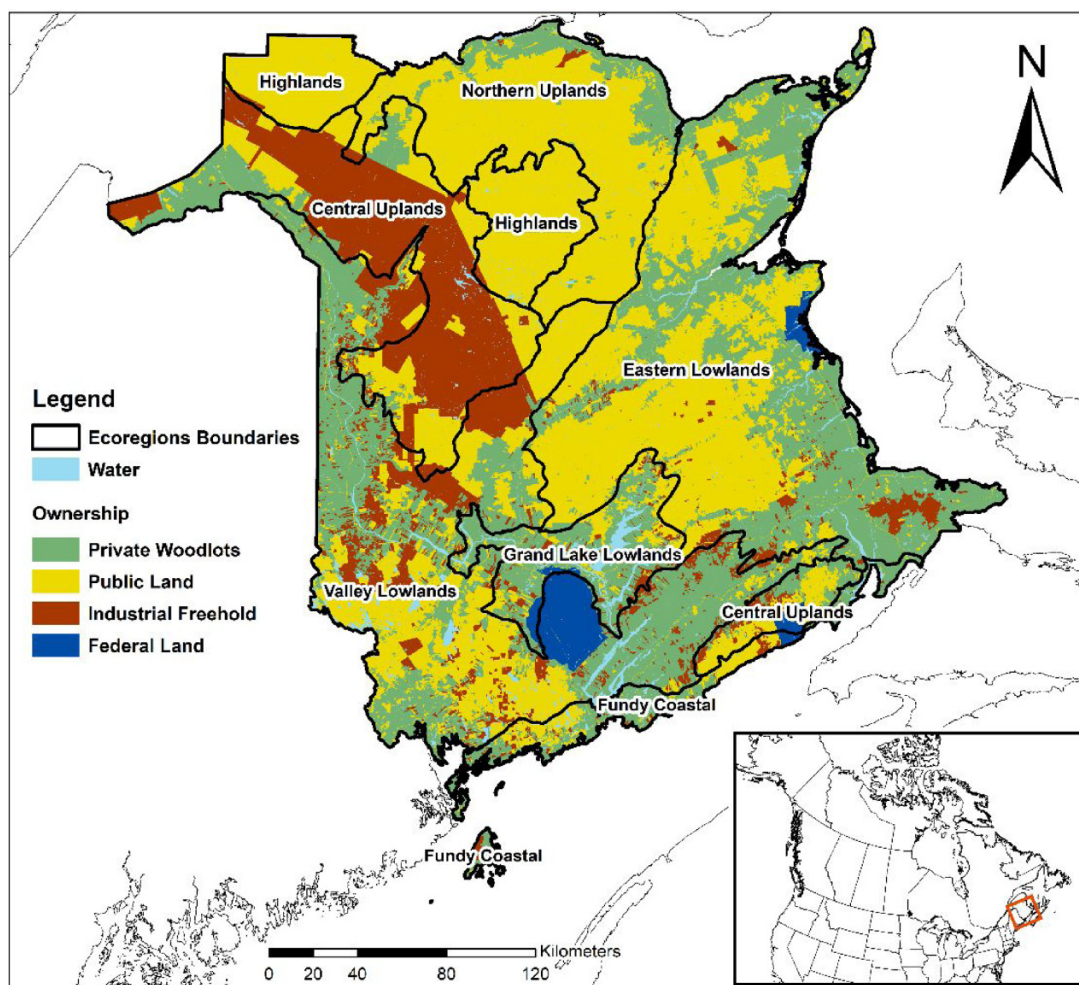
2. Natural history of New Brunswick’s forests

2.1. Climate history and physiography

The physiography of New Brunswick (Fig. 1) was largely shaped by glaciation cycles during the past million years (Zelazny et al. 2007). The region belongs to the Nordic temperate sub-zone of northeastern North America and has a humid continental climate with warm summers and cold winters (averaging 16.5 and −9 °C from 1976 to 2005), with a mean annual precipitation of 1100 mm (1950–2013; Zhang et al. 2019). Local variations in climate conditions are mainly driven by topography and proximity to the ocean (Fortin and Dubreuil 2020). The Atlantic Ocean buffers seasonal temperature variations, with milder winter temperatures and colder summers along the coast. Colder annual temperatures are typically observed in the north-west highlands and uplands of the province compared to other regions (3 and 6 °C),

Dominant natural disturbances that affect New Brunswick’s forests include forest fire, heavy wind, and spruce budworm (*Choristoneura fumiferana* (Clemens, 1865)). On average, from 1885 TO 2021, one hurricane makes land-

Fig. 1. Map of study area location with land ownership provided by the Natural Resources and Energy Development Department of New Brunswick, and ecoregions (from Zelazny et al. 2007).



fall in New Brunswick approximately every 15 years (Taylor et al. 2019). Other forms of wind storms also cause wide-scale windthrow, such as the 1994 nor'easter that affected more than two million cubic metres of wood in the highlands ecoregion (Fig. 1; Bourque et al. 2020). Spruce budworm is the primary insect disturbance in the province (Royama et al. 2005), with outbreaks occurring about every 35 years, and can result in substantial host growth reduction and mortality (MacLean 1980; Johns et al. 2019; Zhang et al. 2020), e.g., 107 million $\text{m}^3 \cdot \text{year}^{-1}$ from 1977 to 1981 (Sterner and Davidson 1982). In addition, several new diseases and insects that were either imported or have migrated into New Brunswick during the last century have caused serious tree damage and increases in mortality rates. For instance, Dutch elm disease (*Ophiostoma ulmi*) has killed over 90% of unprotected elm (*Ulmus* sp.) trees across the province (Magasi and Kanada 1993). The butternut canker (*Ophiognomonia clavigignenti-juglandacearum*) has reduced butternut (*Juglans* sp.) abundance over the last decade with a cankering rate of 60%–75% (Gouvernement of Canada 2013). White pine blister rust (*Cronartium ribicola*) and beech bark disease (*Neonectria faginata*), both imported from Europe in the late 19th century, have infected 3% of white pine (*Pinus strobus* L.) and

99% of American beech (*Fagus grandifolia* Ehrh), respectively (Gouvernement of Canada 2013; Taylor et al. 2013). More recently, the emerald ash borer (*Agrilus planipennis*) that use ash (*Fraxinus* sp.) as its host species has been discovered in the western and southern parts of New Brunswick. This exotic invasive wood borer can cause 100% host population mortality within 8–10 years of arrival (Butler et al. 2022). Due to effective fire suppression efforts, wildfire is relatively uncommon in New Brunswick today (return interval of 3000 years); however, historical records suggest a shorter return interval of 200–600 years was common during the pre-industrial era in New Brunswick (Taylor et al. 2020b). The most recent, significant wildfire event was the great Miramichi fire of 1825 that burned ≈ 1.6 million ha of New Brunswick forests (Stine 2021).

2.2. History of forest management and current forest state

Commercial harvesting started in the early 1700s in the region, as white pine trees were heavily harvested for ship construction (Fellows 1987). As merchantable pines became scarce in the forest, spruce species increased in importance in

the mid-1800s. Both species were then supporting around 600 small sawmills in the province, while eastern hemlock (*Tsuga canadensis* (L.) Carrière) was targeted by the leather industry for the quality of the tannins in its bark (Fellows 1987). As pulp mills were introduced in the province in the early 20th century (e.g., Bathurst Power and Paper in 1915), demand for forest products increased, which led to the harvesting of balsam fir and an increase in the size of the forestry workforce. During this time, hardwoods like yellow birch (*Betula alleghaniensis* Britton) and sugar maple (*Acer saccharum* Marshall) were harvested only for quality lumber and veneer. The first large market for low-quality hardwood only developed with the first hardwood pulp mill construction in 1970; at the same time, sawmills were on a downward trend reaching nearly 100 mills in the late 70s (Holloway et al. 2008). Before and during this period, harvesting levels combined with insect outbreak volume loss exceeded the total growth rate of the forest, leading to a decline in average tree size and remaining old-growth forests (less than 5% of the forested area; Mosseler et al. 2003). From 1976 to 2008, industrial wood use from New Brunswick's forests has risen from 8 to 12 million m³ year⁻¹, following increases in wood imports and a greater utilization of previously marginal species like eastern white cedar and several hardwoods (Erdle et al. 2008). The government adopted the Crown Lands and Forest Act (C-38.1) in 1980 that established 25-year licence agreements for forest companies. The act replaced the hundreds of licences allocated on an ad hoc basis with 10 timber licences each attributed to a licensee (6 licensees remain today managing all 10 licences; Martin et al. 2003). This legal framework established objectives in terms of production but also in relation to forest sustainability, reducing the risk of timber shortage, and provided security to industrial companies.

Today, the forest of New Brunswick is 51% provincial public land (of which 10% is conserved), 29% small private woodlots, 18% large industrial freehold, and 2% federal public land (Fig. 1). The provincial government, through the Natural Resources and Energy Development department, oversees management of public lands, with a harvesting rate of about 1.6% of the total public forest each year (harvest rotation of 60–65 years; NBDERD 2018).

The forest cover area of New Brunswick is approximately 22% balsam fir, 21% black spruce, and 11% poplar (balsam poplar and trembling poplar included) according to the latest forest inventory (2021) made available by the Natural Resources and Energy Development department. Higher-elevation forests, located in the highlands ecoregion, are dominated by boreal species, while low-elevation forests of the lowlands are occupied by a combination of boreal and temperate species (Fig. 1). Hardwood stands, typically dominated by sugar maple, yellow birch, and American beech, are mostly located on rich, upland sites. Pure softwood stands, dominated by balsam fir and red, black, and white spruce, are usually found down valleys and in less well-drained terrains (Zelazny et al. 2007). Red maple acts as an opportunist species and may establish where soils are too acidic to support many other temperate deciduous species or following clearcut disturbances (Fei and Steiner 2009). Black spruce and tamarack (*Larix laricina* (Du Roi) K. Koch) dominate forested wetlands,

while jack pine is mostly located on the dryer and sandier soils of eastern New Brunswick (Zelazny et al. 2007).

Using large-scale mechanical operations, softwood stands (dominated by spruce, fir, and jack pine) are mostly treated with even-age silviculture systems using clearcutting (except in conservation areas where only partial harvesting treatments are allowed). Harvest is followed by planting in approximately one-third of clearcuts on public lands, with the remaining area being naturally regenerated (NBDERD 2018). Early competition within plantations is managed via aerial herbicide application of glyphosate. All plantations, plus 35% of naturally regenerated clearcuts, are treated with pre-commercial thinning around 15–20 years of age following establishment, while around 32% of clearcuts remain untreated (NBDERD 2018). In contrast, hardwood stands (dominated by maple and birch) are typically managed in a way to minimize environmental impacts and ensure natural regeneration. These low-impact treatments are known to include single-tree selection for high-quality stands and shelterwood for lower-quality ones (NHRI 2020). Conducting a single tree selection allows to maintain the overall stand quality, encourages the growth of a diversified understory vegetation, and reduces the risk to natural stressors such as wind, fires, and insect outbreaks (Raymond et al. 2018). On the other hand, less vigorous stands treated with shelterwood help promote the development of a new, healthier, and more diverse forest cohort (Granstrom et al. 2022).

3. Projected climate change

3.1. Climate models, scenarios, and variable selection

We generated monthly climate projections for New Brunswick from an ensemble of 24 daily Canada-wide climate scenarios made available by the Pacific Climate Impacts Consortium and Environment and Climate Change Canada. Downscaling was done using the BCCAQv2 hybrid method, which combines results from Bias Corrected Constructed Analogs (Maurer et al. 2010) and Quantile Delta Mapping (Cannon et al. 2015). The main observational dataset used to calibrate BCCAQv2 was produced by Natural Resources Canada at 300 arcsecond spatial resolution (1/12° grids, ~10 km) over Canada and includes daily minimum and maximum temperature and precipitation amounts for the period 1950–2012 (Hopkinson et al. 2011; McKenney et al. 2011). Monthly data for representative concentration pathways (RCP) 4.5 and 8.5 were derived from daily values and averaged over baseline 1981–2010 and future 2071–2100 periods. RCP 4.5 represents a moderate emission scenario where greenhouse gases emission (GHGE) starts to stabilize by mid-21st century and reaches half of 2050 levels by 2100. On the other hand, RCP 8.5 mimics the continuous rise of GHGE throughout the 21st century, making it the high-emission scenario.

Climate variables were selected to reflect their documented importance in key processes driving forest stand dynamics and health. Temperature and precipitation both exert high control over tree growth, photosynthetic rates, evapotranspi-

ration, respiration, and general tree health. Both variables were modelled for the growing season, where they are most limiting (Babst et al. 2019), which is typically from June to August in New Brunswick. In addition, freeze–thaw events and growing degree-days (GDD) above 5 °C were considered on an annual basis. Freeze–thaw events are useful in predicting frost damage, as they strongly influence the growth of certain tree species like sugar maple (Moreau et al. 2020a, 2020b). GDD are closely related with forest productivity and were determined by calculating the difference between the average daily air temperature and a 5 °C threshold below which biological activity ceases; the differential values are then summed for the entire year (days with an average air temperature below 5 °C have a GDD of zero; Zhang et al. 2019).

By increasing evapotranspiration and vapour pressure deficit, increasing temperatures can exacerbate water deficit and stress in plants despite increases in precipitation (Allen et al. 2010). To estimate the evapotranspirative loss of water with warming, while accounting for changes in precipitation, we calculated a climate moisture index (CMI) using minimum and maximum daily temperatures, precipitation, and elevation data; this drought index is widely used and estimates available atmospheric moisture. In this paper, monthly CMIs for June, July, and August were computed using the Hogg et al. (2013) method of calculation that is based on the “simplified Penman–Monteith” method of estimating potential evapotranspiration (PET). Summer CMI was calculated as the absolute difference between summer precipitation and PET. The CMI is then an indicator of potential growth by estimating the location of wet or dry conditions that can support different types of forests or tree species (Hogg et al. 2013).

3.2. Climate projection results

Summer temperature is projected to increase by 3.5 and 6.2 °C on average under RCP 4.5 and RCP 8.5, respectively, over the 21st century (Fig. 2A); the high-emission scenario temperature prediction corresponds to the current climate in Pennsylvania (NOAA 2022; period 1971–2000). Specifically, the lowland ecoregion, including most of eastern and southern portions of the province except for the cooler Fundy coast, should experience the greatest warming under both RCP scenarios. Winter temperatures (December, January, and February) will increase by 3.8–6.6 °C on average. Southern regions of New Brunswick will cope with winter average temperatures above the freezing point by the end of this century, while highland winters will remain relatively cool (–7 to –4 °C).

Increasing temperatures are projected to enhance GDD by 39%–77% from ≈1700 to 2360–3000 degree-days on average by the end of the 21st century (Fig. 2B); 3000 degree-days roughly correspond to the 2019 values recorded in Ohio, USA. In New Brunswick, the Grand Lake lowland ecoregion is projected to achieve the highest GDD values for each RCP scenario (2800–3400 GDD). For highland ecoregions, values will range between 2200 and 2600 GDD in both RCP scenarios for the projection period. However, the magnitude of change is similar for all New Brunswick regions.

Despite these large projected increases in temperatures, summer precipitation is expected to display only moderate increases of 5%–9% (13–25 mm) under the different RCP scenarios relative to the baseline summer precipitation of approximately 278 mm (Fig. 2C). In comparison, winter precipitations should increase significantly more, with 9%–31% increases (25–83 mm) relative to current levels. In the highlands ecoregion of the province, rainfall during summer is projected to stay relatively stable (325–350 mm), contrary to the lowlands, central uplands, and Fundy coastal ecoregions, which are projected to experience significant increases in winter precipitation, from the baseline of 250–275 mm to projected 350–400 mm. Across New Brunswick, the mean annual precipitation will increase from baseline 1200 mm to 1250–1450 mm. The Fundy coastal ecoregion is expected to experience wetter annual conditions under RCP 8.5 (1450–1700 mm) compared to surrounding areas due to its proximity to water.

The projected increases in summer precipitation will be insufficient to counteract the negative effect of future warming on available moisture, expressed here as the CMI. For the 2071–2100 period, summer available moisture (CMI) is anticipated to decrease by 48–79 mm across RCP scenarios, because of the large summer warming (+3.5–6.2 °C) coupled with only moderate changes in summer precipitation (+13–25 mm; Fig. 2D). The lowland ecoregions are projected to display the highest atmospheric water deficits of 60–150 mm under various RCP scenarios, while uplands and the Fundy coast will encounter limited water deficit (CMI between –30 and +60 mm) during the summer period.

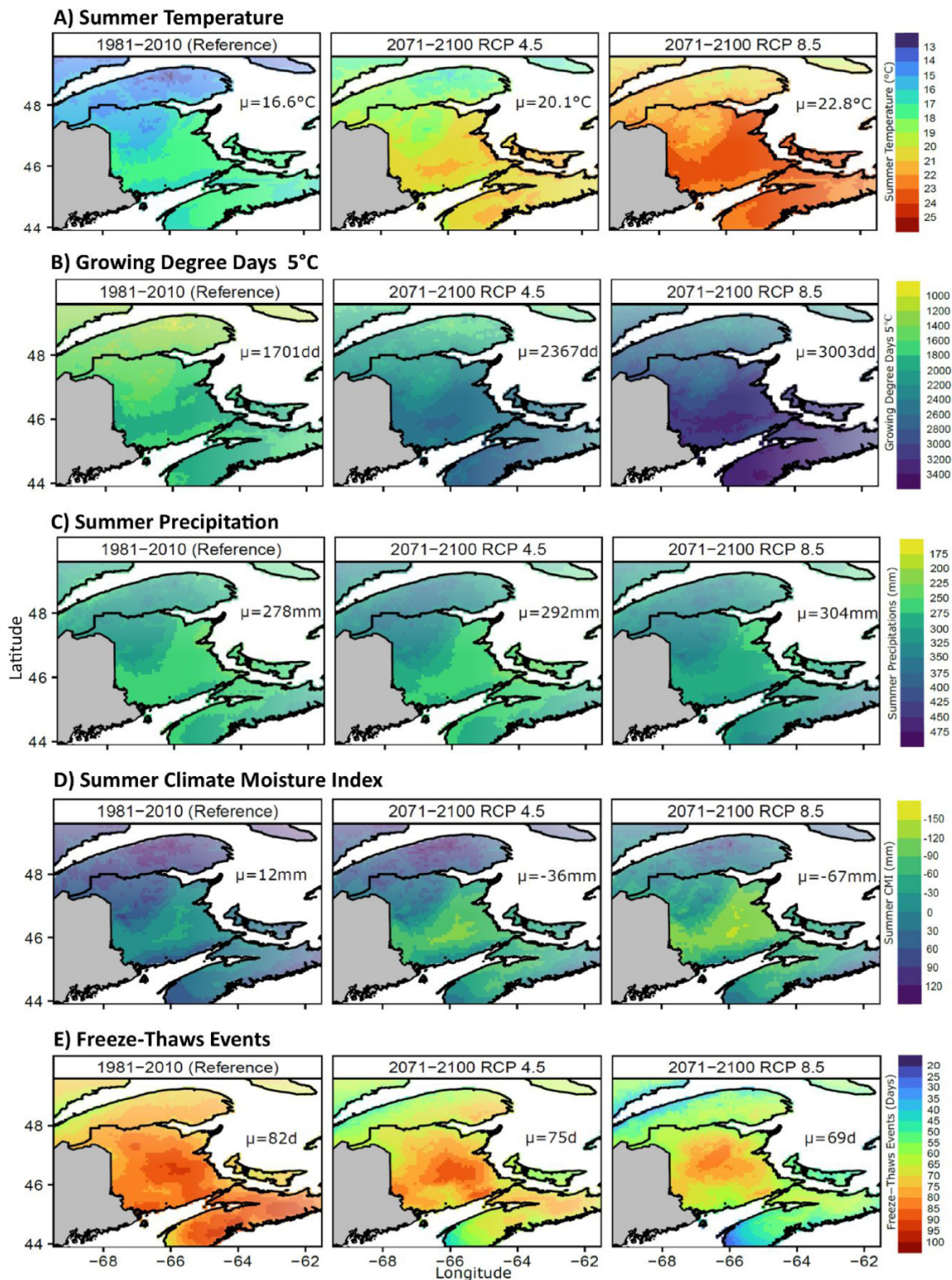
Freeze–thaw events, defined here as the number of days when air temperature fluctuates above and below 0 °C, should not increase in the future. In fact, freeze–thaw events are projected to display a marginal decrease in frequency by the end of the 21st century, from 82 days (baseline) to 79–69 days in 2071–2100 (Fig. 2E). The lowlands ecoregion of New Brunswick that is currently more susceptible to freeze–thaw events should remain equally susceptible under RCP 4.5, but radiative forcing under RCP 8.5 projects that the area of vulnerability will shift toward the central highlands of the province. Northeastern New Brunswick will be least affected, while the southern part of the central uplands ecoregion will display the highest occurrence of freeze–thaw events.

4. Climate change impacts on New Brunswick forests

4.1. Direct effects on forest composition and productivity

Forest tree species are estimated to have had a maximum natural migration velocity of 0.5 km·year^{–1} historically following the retreat of the last glaciation, which is slower than the 3–5 km·year^{–1} estimated to keep pace with climate change-induced displacement of their adequate climate zones, leading to maladaptation (Sittaro et al. 2017). Most simulation models of forest dynamics agree that boreal tree species abundance will substantially decline to the benefit of temperate species under climate change in New Brunswick,

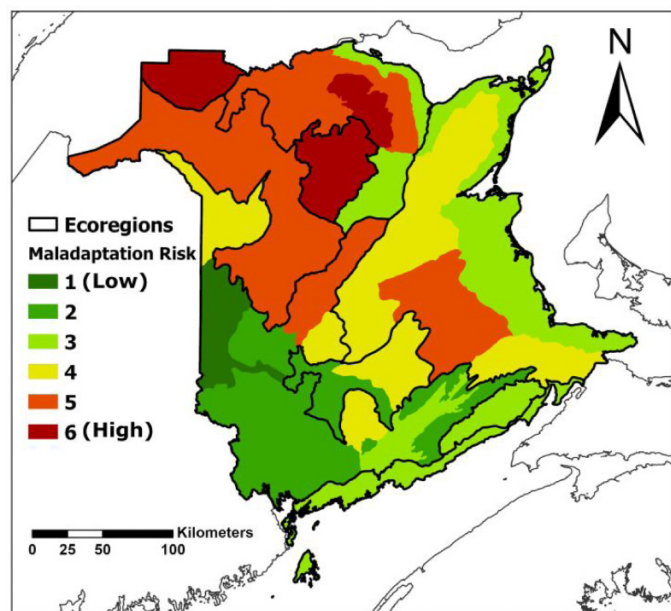
Fig. 2. 30-year average projections of (A) summer temperature, (B) growing degree-days 5 °C, (C) summer precipitation, (D) summer CMI, and (E) freeze-thaw events. Baseline period 1981–2010 (left), RCP 4.5 2071–2100 (centre), and RCP 8.5 2071–2100 (right). Summer months include June, July, and August, while average values are for New Brunswick only.



as the southern edge of their range shifts northward (Fig. 3; Taylor et al. 2017; D'Orangeville et al. 2018; Bouchard et al. 2019; Brecka et al. 2020; Boulanger et al. 2021). More specifically, the relative abundance of balsam fir and black spruce is

projected to decline by as much as 50%, while jack pine may decline by 25% by 2100 under RCP 8.5 (Taylor et al. 2017). The rate of decline is projected to be marginal at first, but then gradually increase by 2050, facilitating increases in the rela-

Fig. 3. New Brunswick forest maladaptation risk based on the relative abundance of tree species projected to decline under climate warming (balsam fir, black spruce, white spruce, tamarack larch, jack pine, and white birch), modified from Zelazny et al. (2007). Note: maladaptation risk 1 = <10%, 2 = <20%, 3 = <30%, 4 = <40%, 5 = <50%, and 6 = >50% in relative abundance.



tive abundance of temperate tree species. Notably, red maple is expected to double its relative abundance to represent 30% of the New Brunswick forest composition by 2100 (Taylor et al. 2017; Brecka et al. 2020). American beech (+135%) and white pine (+35%) are also projected to display large increases in abundance under similar scenarios during the same timeline (Taylor et al. 2017), although the long-term effect of ongoing beech bark disease remains poorly understood. The fate of other temperate species like sugar maple is unclear. Some studies project limited changes in sugar maple distribution over the next century (Steenberg et al. 2013; Zolkos et al. 2015), while others predict a decline due to increasing competition by American beech (St-Jean et al. 2021) and climate anomalies (Moreau et al. 2020a). For yellow birch, model projections for the Acadian Forest Region range from steady decreases starting in 2040 (Bouchard et al. 2019) to 25% increases (Taylor et al. 2017). However, models that account for harvesting impacts under climate change predict significant declines in yellow birch abundance (Steenberg et al. 2013; Boulanger et al. 2021). The general shifts in abundance are considered largely the result of changes in the competitiveness between tree species, thus modifying demographic rates that have strong control over the success or failure of a species establishment (Swenson et al. 2020).

Among the few studies projecting climate change impacts on tree species composition in the New Brunswick forests, Taylor et al. (2017) used an individual tree-based model, PICUS, which projected a 6%–42% decline in boreal species

growth by the end of the 21st century under the RCP 4.5 and 8.5 climate forcing scenarios. In contrast, the same study predicts stable growth rates of temperate species until around 2050, which then increases drastically during the 2050–2100 period under RCP 8.5, shifting from an average of $4\text{--}7\text{ m}^3\cdot\text{ha}^{-1}\cdot\text{year}^{-1}$. The temperate species growth increase is mainly driven by hardwood species American beech, red maple, red oak, and white ash (*Fraxinus americana* L.), when no disturbances are considered. Overall, moderate warming of New Brunswick could be beneficial to forest productivity until the mid 21st century, when gains in growth of temperate species will no longer offset growth reductions in boreal species resulting in a net decline in forest productivity (D'Orangeville et al. 2018; Molina et al. 2021).

The boreal species that are projected to decline under climate change are not equally distributed across New Brunswick (Fig. 3). The highlands as well as the central and northern uplands ecoregions will be most severely impacted, as 45%–55% of their landscape is covered by balsam fir and spruce species (Fig. 3). Eastern lowlands display moderate risks in general, but higher black spruce components increase the maladaptation risk in some locations (Fig. 3). Balsam fir and white and black spruce, considered at-risk species, are in lower abundance in south-western New Brunswick, where sugar maple, beech, aspen, and yellow birch are more common, and at-risk species represent less than 20% of total forest cover (Fig. 3).

4.2. Indirect effects of climate change from natural disturbances

Evidence suggests that the frequency and severity of some natural disturbances will increase over this century by up to 10%–15% relative to the 1981–2021 baseline by 2100; wind, fires, and drought are expected to deal the biggest damages, while spruce budworm dieback decreases steadily south of its range as climate warms (Boucher et al. 2018; Taylor et al. 2020b). Brecka et al. (2020) conducted a wide-scale study projecting wood supply under different climate change scenarios using the landscape model LANDIS-II in eastern and central Canada. In all scenarios and locations examined, forest biomass lost to fires and drought increased over the 21st century, especially under RCP 8.5, to a point where biomass lost to harvest and natural disturbance exceeded overall forest growth. This threshold was reached sooner in central Canada where tree diversity is lower, fire disturbance rates are higher, and drought is more severe compared to eastern Canadian forests.

Drought is associated with severe growth reductions that can impact carbon storage, wood supply, and wildlife habitat (Brzostek et al. 2014; D'Orangeville et al. 2018). Drought-induced tree mortality is less common in eastern Canada compared to drier regions of central Canada, where drought-related forest mortality has increased $4.7\%\cdot\text{year}^{-1}$ in recent decades (Peng et al. 2011). Sánchez-Pinillos et al. (2022) suggests that drought-induced mortality will increase in eastern Canada with higher frequencies of low-intensity drought. Moreover, the negative effects of drought can last

for many years (Scharnweber et al. 2020) and predispose trees to other stressors like insects and diseases (Anderegg et al. 2015; Broom 2023).

Increasing temperatures and moisture deficits should also exacerbate fire risks (Boulanger et al. 2013; Boucher et al. 2020). Not accounting for historical and future fire suppression efforts, the area burned across New Brunswick should increase from a baseline (1990–2021) of 1000 ha·year⁻¹ to 2000–3000 ha·year⁻¹ (Boucher et al. 2020; CCFM 2021). In southern and northern New Brunswick, where conditions are warm and dry, the annual area burned is projected to be four and two times higher than current level, respectively (Boulanger et al. 2013). However, these increases could be partially offset by fire suppression efforts as well as the projected increase in the proportion of hardwoods, which can reduce fire propagation rates by up to 50% in these areas (Terrier et al. 2013; Boulanger et al. 2017).

Spruce budworm is also a major natural disturbance agent of spruce and fir forests in eastern North America, including the Acadian forest (MacLean 2016). Climate change is projected to shift the range of spruce budworm northwards (Taylor et al. 2020b). Outbreaks in the extreme northern part of New Brunswick are projected to lengthen by 2–4 years and result in 15% more defoliation in 2081–2100 period relative to the 1981–2010 period (Gray 2013). To the contrary, warming could reduce the severity of future infestations by 40%–50% (7–10 year reduction in severe defoliation) in southern New Brunswick by mid-21st century (Gray 2013; Pureswaran et al. 2019), but uncertainties remain due to expected shifts in spruce phenology with warming (Bellemin-Noël et al. 2021). Other insects and pathogens, historically constrained to the south of New Brunswick by cold climate, such as hemlock woolly adelgid (*Adelges tsugae* Annand), may migrate north with warming, thereby facilitating the establishment of novel forest pest species (DeSantis et al. 2013; Russo et al. 2019).

Wind events are important drivers of gap-phase stand dynamics in New Brunswick forests, but can also cause major windthrow, producing significant tree mortality and increasing forest management complexity (Klaus et al. 2011; Taylor et al. 2020b). Wind events can be classified into two main categories: (1) tropical cyclones, characterized by sustained wind speeds of at least 104 km·h⁻¹, and (2) windstorms, encompassing a wide range of non-hurricane winds such as extratropical cyclones and convective storm cells. Among these categories, tropical cyclones inflict the most significant damage to forests due to their sheer force and extensive spatial and temporal coverage, in comparison to non-hurricane storms. However, these events are relatively rare, occurring approximately once every seven years in Atlantic Canada, with approximately 25% of the affected area experiencing severe windthrow (Taylor et al. 2019, 2020b). On the other hand, convective storm cells cover relatively smaller areas ranging from tens to hundreds of square kilometres, resulting in windthrow events affecting small, dispersed patches of 100–1000 m² or even extending over hundreds of square kilometres. Now that suppression of fire and insect outbreak control is effective, wind has become the dominant disturbance in Atlantic Canada.

Among Acadian forest tree species, balsam fir is one of the most susceptible to windthrow due to its shallow rooting habit, weak anchoring, and also its crown and branching architecture that acts as a sail (Ruel 2000; Taylor et al. 2019). While wind speeds are forecasted to increase only marginally (<3%), projected soil warming may reduce the anchoring benefit of frozen soils, leading to an increase from 3% to 30% of balsam fir stands impacted by windthrow (Bourque et al. 2020). However, increase in windthrow risk may be offset by the reduction in balsam fir abundance projected by the end of the 21st century, leading to an overall decline in windthrow (Saad et al. 2017).

5. Resiliency indicators and management solutions for climate change

The sustainable production of goods and services to meet society's needs is central to the objectives of forest management. However, increasing risks and uncertainty associated with climate change is contributing to complexity of this already complex process. Promoting climate change-resilient forest landscapes requires a good understanding of stand attributes that mitigate these risks (e.g., Dymond et al. 2015; Bryant et al. 2019). In this section, we attempt to summarize the forest landscape attributes that have been linked with resiliency to climate change and propose stand interventions to maximize them. We acknowledge that the current state of knowledge of these indicators in New Brunswick is limited, which prevents the development of quantitative thresholds in stand attributes that have been identified to promote resilience and can be integrated into forest planning models to manage for climate change.

5.1. Resilience indicators for climate change

Diversity in forest tree species plays a critical role in enhancing the capacity of stands to cope with changing climatic conditions and disturbance regimes (Jalonen et al. 2018; Nef et al. 2021). For instance, during insect outbreaks, more diverse stands have been observed to experience reduced defoliation compared to less diverse stands (Hisano et al. 2018). This applies not only to species diversity but also to intraspecific genetic diversity, which positively impacts productivity, natural regeneration success, and resistance to pests (Tito de Moraes et al. 2020). Natural forests with higher species diversity have also demonstrated superior resistance to disturbance, offering enhanced resilience against external pressures (Table 1; Jactel et al. 2017). Furthermore, at the landscape level, diversity in stand structures and composition acts as an insurance against climate change impacts (Messier et al. 2019).

Projected increases in fire risk, resulting from higher temperatures and dryer conditions, also present challenges for forest resilience (Boulanger et al. 2013; Wang et al. 2015b). Over 95% of fires in Atlantic Canada start by human ignition (Taylor et al. 2020b). If forest management practices are not adapted and foresters are not careful when working in the woods, this may lead to some decrease in for-

Table 1. Resilience indicators for three disturbance agents frequently occurring in the Acadian forest.

Disturbance agent	Indicator	Resilience effect
Fires	Area of 0–29 years old forest of all types (ha)	↑
	Area of 30+ years old pure deciduous and mixed-deciduous (conifer < 50%; ha)	↑
	Area of 30+ years old pure coniferous and mixed-coniferous (conifer > 50%; ha)	↓
Insects/pathogens	Tree species diversity within a stand	↑
	Area of monospecific stand (>80% of basal area occupied by the same species; ha)	↓
	Volume of susceptible species to insects and pathogens (m ³)	↓
Drought	Area thinned (40%–70% basal area removed) with time since thinning under 40 years (ha)	↑
	Abundance in drought-tolerant species, including eastern white pine, red oak, white ash, and red spruce (ha)	↑
	Ratio of total forested areas (ha) over total area clearcut (ha) younger than 15 years (ratio)	↑

Note: ↓, lower is better; ↑, higher is better; ha, hectare; m³, metres cube.

est productivity and higher fire suppression costs (Table 1; Hope et al. 2016; Janowiak et al. 2018). The flammability of stands varies with stand characteristics, as hardwood-dominated and young coniferous stands (0–29 years) generate less flammable material compared to mature coniferous stands (30+ years of age; Bernier et al. 2016). Conifer-dominated stands also experience two-to-five times higher fire occurrence, posing risks of understocking and regeneration failure, particularly for boreal species at their southern distribution limit (Stevens-Rumann et al. 2018; Hart et al. 2019). Conversely, certain species like white pines, with a broader distribution south of New Brunswick, may exhibit different responses.

Promoting drought-adapted species is crucial for building climate-resilient stands and forests. Species such as oaks, characterized by high hydraulic resistance to embolisms, display increased competitiveness and growth even during drought conditions (McDowell et al. 2008). Some tree species, including white ash, eastern white pine, and jack pine, exhibit an earlier growth phenology, allowing them to concentrate growth during periods of abundant water early in the growing season (Table 1; D'Orangeville et al. 2022; Pearson and D'Orangeville 2022). At the stand level, potential drought effects can also be temporarily mitigated through lower crowding, which allow the redistribution of available water to a reduced number of trees. Commercial thinning trials in a 26-year-old Norway spruce (*Picea abies* (L.)) plantation lowered stand-level transpiration by 25%–50% (Gebhardt et al. 2014), increasing tree-level transpiration by 30%–50%, an indicator of high photosynthesis and growth. However, the beneficial effects of thinning are limited in duration, typically lasting 20–40 years post-treatment, as larger trees in thinned stands become more sensitive to drought (Table 1; Manrique-Alba et al. 2022). Clearcutting and clearing areas also have implications for water loss, reducing soil water holding capacity and retention through the reduction of organic material, elevated soil temperatures, and lower transpiration rates (Bawafi et al. 2020). The effects of such disturbances on water runoff and retention can persist for 10–15 years, varying with seasonality (Ide et al. 2013).

5.2. Forest interventions influencing resilience to climate change

Current empirical knowledge on the effect of silvicultural interventions on forest resilience remains very scarce. In response to this knowledge gap, several adaptive silviculture trials are underway in North America. Notably, the Adaptive Silviculture for Climate Change research project is establishing a network of silvicultural trials at several locations in the US and one site in Petawawa, Canada (Nagel et al. 2017). The trials aim to compare the results of status-quo silvicultural prescriptions with novel silvicultural regimes under climate change uncertainty to guide the forest toward three different management states based on resistance, resilience, and transition (Nagel et al. 2017). These concepts provide a framework for managing forests in the context of climate change by focusing on minimizing impacts (resistance), enhancing recovery and adaptability (resilience), and facilitating shifts in composition toward more climate-resilient states (transition). Another project, the Resist-Accept-Direct framework was developed by National Park Services in the US to assist managers in making informed choices about how to respond to a climate-induced forest trajectory change (Schuurman et al. 2020). These research avenues all have in common adopting of new approaches and monitoring methods to increase the adaptability of forested ecosystems to changing climate through increased diversification, complexity, and capacity for self-reorganization of the forest (Nocentini 2015).

Building upon the current knowledge gap surrounding the effects of silvicultural interventions on forest resilience, it is of importance to explore and comprehend the potential of various intervention methods. Thinning, shelterwood and two-age cover harvest, assisted migration, and clearcutting stand as prominent intervention methods have garnered considerable attention from both forest industries and scientific researchers (e.g., Roy and Nolet 2018; Taylor et al. 2020a; Pelai et al. 2021; Moreau et al. 2022). Understanding the benefits, limitations, and considerations associated with each intervention method holds the key to gaining valuable insights into their efficacy and their potential contributions to fostering resilience in forested landscape (Messier et al. 2019). By delving into these interventions and critically evaluating

their implications, we can pave the way towards sustainable forest management practices that optimize the resilience of forested landscapes.

5.2.1. Thinning

Forest thinning is a tending treatment that entails the strategic removal of trees during their early developmental stages to mitigate the intensity of competition among the remaining trees. By selectively removing certain trees, forest managers aim to guide and manipulate the composition of the stands, with the general goal of improving the overall quality of the growing stock (Ashton and Kelty 2018). Thinning has many other benefits ranging from increased stand productivity to higher resilience (Table 2). Among its main benefits, thinning can decrease fire risk by reducing fuel loads (Jain et al. 2020), enhance tree performance by reducing competition for resources such as water, light, and minerals (Navarro-Cerrillo et al. 2019; Knapp et al. 2021), increase tree vigour to survive insect outbreaks (Hood et al. 2016), and stimulate root development to diminish the chance of windthrow (Picchio et al. 2020; Table 2). However, thinning treatments should be used with caution, as poor species selection or inadequate stand structure may result in adverse effects where the risk of natural stressors becomes greater, resulting in increased damage; Table 2). The main limitation of thinning is that the treatment temporarily increases the risk of wind damage to residual trees, which can worsen for older stands and tree height (Moreau et al. 2022), meaning that thinning performed in older stands without previous treatments that aim at solidifying the structure should be avoided.

Studies examining thinning effects under climate change have primarily focused on the impacts of individual disturbance agents, such as wildfires, insect outbreaks, wind, or droughts (e.g., Sohn et al. 2016; Achim et al. 2022; Rabin et al. 2022; Table 2). However, emerging research in the field of natural disturbance ecology suggests that under climate change, these disturbance agents are increasingly likely to interact with one another (Taylor et al. 2020b; Ruel et al. 2023). This interaction can result in complex and synergistic effects on forest ecosystems that influence the beneficial effects of thinning treatments (Moreau et al. 2022), effects that are still relatively unknown. Future research should explore the specific mechanisms and processes through which disturbance interactions influence thinning effectiveness, providing valuable insights for sustainable forest management in a changing climate.

5.2.2. Shelterwood and two-age cover harvesting

Shelterwood harvesting is a silvicultural method used to promote forest regeneration while maintaining ecosystem integrity. This partial regeneration harvest intervention involves a series of partial cuts spread over a short period, leading to the establishment of an even-aged cohort of regeneration while preserving canopy cover during the establishment phase (Ashton and Kelty 2018). The treatment

typically comprises four phases: preparatory cut, seedling establishment cut, secondary cut, and final cut (Boulet et al. 2013). The preparatory and seedling establishment cuts are commonly combined in the first harvest, while the secondary and final cuts are usually grouped in the final harvest to optimize treatment costs (Boulet et al. 2013).

Shelterwood harvesting offers several advantages over clearcutting, making it an ecologically valuable forest management practice. By balancing timber production with species and structural diversity, it contributes to maintaining a higher variety of tree sizes, including dead trees, which are key to enhancing resilience against climate change (Kohm and Franklin 1997; Fenton et al. 2013). The presence of dead trees within the canopy provides habitat diversity and food availability for numerous faunal species, supporting overall ecosystem health (Kohm and Franklin 1997). Furthermore, shelterwood harvesting has been shown to foster vigorous and dense regeneration compared to clearcut sites, as evidenced in a Czech Republic experimental study conducted in even-aged Scots pine (*Pinus sylvestris* L.) stands (Brichta et al. 2020). This characteristic of shelterwood harvesting makes it a valuable tool in promoting forest regeneration and recovery. In addition to its positive impact on biodiversity and regeneration, the progressive nature of the shelterwood regeneration process aligns with five out of the six principles of climate change adaptation, as identified by Brang et al. (2014). Moreover, shelterwood harvesting has proven effective in managing invasive species, such as the ongoing American beech invasion of shade-tolerant stands in eastern North America (Roy and Nolet 2018). The treatment is restoring the balance between tree species and mitigating the adverse effects of invasions (Roy and Nolet 2018). Further, shelterwood harvest systems that retain at least 50% of the initial basal area buffers environmental stress on seedlings, increasing their survival during the first years of establishment (Mallik et al. 2013).

The two-age continuous cover management system shares similarities with shelterwood, as both approaches emphasize maintaining canopy cover during regeneration. However, the two-age continuous cover system is distinctive in its focus on always sustaining two distinct age classes (NHRI 2020). Its growing popularity stems from the continuous supply of forest products while preserving the protective canopy benefits, buffering regeneration from external climatic conditions (Harrington et al. 2022). Being a continuous cover method, this approach offers unique forest values with its more complex vertical and horizontal structure, a feature that cannot be solely provided by shelterwood or clearcut harvests, positively affecting resilience over the landscape treated with a mosaic of these different treatments (Bose et al. 2020; Girona et al. 2023).

The implementation of shelterwood and two-age cover harvests in forest management has demonstrated numerous benefits, yet it is essential to acknowledge and address potential drawbacks associated with these approaches. The establishment of an even-aged regeneration cohort may be hindered by inadequate canopy gaps, limiting successful natural regeneration and potentially affecting forest biodiversity and the rate of success of the treatment (Bose et al. 2023). More-

Table 2. Overview of benefits, objectives, and potential problems associated with the application of thinning in the event of distinct disturbances and in the context of climate change.

Disturbance	Benefits and objectives	Potential problems	Reference
Fire	• Intensive thinning reduces fire risks by harvesting mortality and reducing fuel availability in future years • Reduce negative effects of severe fires • In case of fire, treated areas have a more homogenous fire behaviour	• Rise in forest floor fuel availability immediately after thinning increasing fire risks in the short time • A complementary understory cleaning is needed to enhance effect on reducing fire risks • High crown/height ratio could in some forest types increase fire severity of crown burn	Hevia et al. 2018; Piqué and Domènech 2018; Jain et al. 2020
Drought	• Intensive thinning enhances tree performance and distributes available water to a smaller number of individuals, increasing water use efficiency	• Increase in drought risks if only one treatment is conducted when natural mortality has been absorbed by remaining trees, increasing tree size and water demand	D'Amato et al. 2013; Navarro-Cerrillo et al. 2019; Knapp et al. 2021
Insect outbreaks	• Increased tree vigour reduces tree vulnerability to insect attacks and diminishes the risk of death in the case of an attack • Reduced density limits the spread rate of minor to moderate outbreak to other stands • Control residual species composition to reduce infestation risks	• Defoliation can be higher in thinned stands but high leaf area due to enhanced light conditions counterbalances the effect of elevated defoliation • Thinned stands that have not yet responded to the treatment are particularly at risk	Ostaff et al. 2006; Hood et al. 2016; Rossi et al. 2018
Wind	• Thinning occurring at an early developmental stage will stimulate root development, reducing wind damage and windthrow risks • Thinning from below has a reduced risk of wind damage	• Late thinning usually destabilizes stands because of the high wind exposure of residual trees with high height/diameter ratio • Vulnerability due to late thinning can last up to 10 years in less productive stands • Wind damage susceptibility improves with tree height and stand age at thinning	Picchio et al. 2020; Torita and Masaka 2020; Duperat et al. 2021

over, the retention of residual trees could create favourable conditions for pests and diseases, posing challenges to stand health if not alleviated by high landscape biodiversity (Bose et al. 2023; Girona et al. 2023). On the other hand, partial harvests demand intricate management, leading to higher operational costs and reduced timber output during selective harvesting (Soman et al. 2019). The success of these methods and the level of resilience gained are highly dependent on the selection of appropriate trees at the time of harvest and the extent of damages to residual stems (Moussaoui et al. 2020).

5.2.3. Clearcutting

Clearcutting, a logging treatment involving the complete removal of trees in a specific area, has received significant attention in forest management strategies amid the context of climate change, primarily due to its potential implications for carbon sequestration, forest regeneration, biodiversity, and its indirect impact on conservation. In the specific context of New Brunswick, clearcutting's effects on regeneration are variable, with a tendency to favour fast-growing and shade-intolerant species (Taylor et al. 2020a). Notably,

red maple, white birch, and poplars have been identified as the most opportunistic merchantable tree species, although birches and poplars face challenges due to their boreal traits in the changing climate (Taylor et al. 2017). Clearcutting is particularly suitable in situations where natural regeneration is limited or when certain tree species require open spaces for growth that cannot be achieved through alternative treatments (Ashton and Keltý 2018). Furthermore, clearcutting, followed by reforestation, results in the establishment of younger, more vigorous forests with a higher potential for carbon sequestration compared to older, mature forests. Rebane et al. (2020) demonstrates that naturally regenerated boreal forests following clearcutting transition from being carbon sources to carbon sinks within 5–6 years post-harvest. Additionally, clearcutting can aid in the control of pest and disease spread, thereby safeguarding the health of adjacent forest stands (Hartshorn 2021; Tsikas and Karanikola 2022). Importantly, the intensive nature of clearcutting, when followed by plantations, indirectly contributes to the conservation of forest areas by concentrating harvest activities in limited portions of the landscape (Côté et al. 2010; Himes et al. 2022). This land zoning approach,

akin to those being implemented in Nova Scotia and Quebec (Thuot et al. 2005; Lahey 2019), holds promise for enhancing conservation efforts in the region.

However, it is crucial to acknowledge the potential drawbacks and negative impacts associated with clearcutting, particularly in the context of climate change, as highlighted in various studies. Clearcutting is generally regarded in the literature as a transitional treatment rather than one that strictly promotes forest resiliency (Bataineh et al. 2013; Ibarra et al. 2017; Nagel et al. 2017). First, clearcutting eliminates the protective cover provided by trees, exacerbating the negative effects of climate change-induced heat and water stress (Boros et al. 2019; De Lombaerde et al. 2022). This practice tends to simplify the species composition and structural diversity of the forest, resulting in even-aged monospecific stands, especially when plantations are established (Kusumoto et al. 2015). At the landscape level, clearcutting is likely to contribute to habitat fragmentation, jeopardizing numerous species that rely on complex forest structures and diverse microhabitats, which are lost through clearcutting (Lehmkuhl and Ruggiero 1991; Fortuin and Gandhi 2021). Furthermore, the complete removal of on-site trees eliminates the root systems that play a crucial role in soil stabilization, leading to increased soil erosion and a subsequent decline in soil fertility for subsequent cohorts (Fielding et al. 2022).

In New Brunswick, planted clearcuts represent around 33% of clearcut areas and undergo mechanical site preparation followed by the plantation of predominantly white and black spruce (NBDERD 2018). It is worth noting that these two species have been found to exhibit low tolerance to future warming (Taylor et al. 2017). The response of planted trees to climate change may differ from that of natural stands and populations due to factors such as vegetation control and the superior growth traits of the planted trees, which have been developed through extensive tree breeding programs (Ennos 2015). Furthermore, plantations generally have significantly shorter rotation periods compared to natural forests, which reduces their vulnerability window to natural stressors and, consequently, their susceptibility to climate change (McEwan et al. 2020). However, there are important knowledge gaps concerning how well the white and black spruce genotypes in these plantations can respond favourably to climate change under tree improvement programs.

5.2.4. Assisted migration

Assisted migration is the human-assisted movement of tree species and populations in response to climate change. The practice consists of matching climates in which seed sources evolved with near future climates for plantation sites (Pedlar et al. 2011). It is also a key silvicultural practice to help tree species that are threatened by rapid climate change to overcome reductions in regeneration and growth, and increasing mortality attributable to climate changes (Löf et al. 2019; Barrette et al. 2021). Assisted migration is used to artificially speed up the migration of warm-adapted populations, to increase species diversity, and to promote better adapted species, ensuring a sustainable future productivity in

the managed forest (Pelai et al. 2021). One potential challenge of assisted migration is the uniqueness of each species regarding their local adaptation and phenotypic plasticity but also the capacity of genotypes to not only tolerate future, warmer conditions but also current, colder conditions, making assisted migration risky and research-intensive (Sáenz-Romero et al. 2021). In the context of the 21st century climate change projections, assisted migration strategies will have to be implemented based on species-specific vulnerabilities to environmental changes and industrial needs (e.g., spruce sp. in New Brunswick). Consultations with first nations are important when implementing assisted migration mainly because first nations possess rich traditional ecological knowledge that encompasses a deep understanding of local ecosystems that have a high chance of enhancing the effectiveness and success of assisted migration efforts (Berkes et al. 2000). Collaboration with these nations creates opportunities for shared decision-making, co-management, and mutual learning while respecting the deep cultural connections that trees are playing in their traditional practices (Whyte et al. 2019).

Currently, assisted migration is mostly being used and tested only for research in eastern Canada; the industrial implementation of this practice is still pending in almost all locations due to research and seed source limitations (Klenk Larson 2015). Some species are more responsive to assisted migration than others because they are less sensitive to warming and therefore have a better survival rate in a colder environment when being migrated. Typically, the natural distribution range of species is a good indicator of such ability and tolerance to climatic change; widespread distribution ranges show an ability to develop under a wide range of climate conditions (e.g., red spruce vs. red oak; Williams and Dumroese 2013; Guan et al. 2021). Some assisted migration trials currently in place in western Canada reveal that the plantation of white spruce (*Picea glauca* Voss.) from seed sources 3 °C warmer than plantation sites can mitigate the negative impact of climate change on productivity (Sáenz-Romero et al. 2021). Similar trials are conducted in Ontario and Quebec, where southern seed provenances are tested to be used further north (e.g., Lu et al. 2014; Otis Prud'homme et al. 2018). Species ecological niches must be carefully studied before a species is planted and introduced into a new region to avoid large-scale regeneration failure from different factors like frost hardiness, competition, predators, etc. (Park et al. 2014; Sebastian-Azcona et al. 2019). It is important to note that southern population migration can lead to “decoupling” with seasonality in migrated regions increasing the risk of early leaf unfolding leading to spring frost damage (Leites et al. 2019). Furthermore, deer and moose browsing susceptibility might be an important criterion in species selection given its influence on seedling survival in eastern Canada (Champagne 2021).

6. Conclusion

Forest conditions and productivity will be significantly altered by a 6 °C increase in summer temperatures and increasingly negative CMI, especially at the end of the current cen-

tury. This is happening as public demand for woody material and expectations toward conservation are higher than ever. In New Brunswick, cold-adapted northern species populations will decline, while warm-adapted temperate species will thrive. Adaptive silviculture for climate change programs are suggesting various management techniques that promote forest resilience to climate change, and several trials and management efforts are already in place through the northern hemisphere. Meanwhile, resilience indicators are being developed to accurately monitor the capacity of ecosystems to cope with the range of predicted changes and to fulfil human demand for forest products and services. Thinning, shelterwood and two-age harvest, and assisted migration are important silvicultural tools identified by adaptive programs to use to create resilient forest conditions and temper climate change negative effects. However, most agencies must develop the ability to incorporate climate change and resiliency into planning and prescription development. On the way forward, foresters and scientists need to focus on the integration and the understanding of multi-interacting stressors and management strategies rather than concentrate on single events and solutions to understand how we can include constant environmental changes in our planning for sustainable forest management. Foresters have just begun their adaptation to this new reality, and this review is gathering tools that are essential to continue the implementation of multi-scale climate adaptive forest management.

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Data availability

Climate data generated or analyzed during this study are available from Ouranos Climate Centre. Data are however available from the authors upon reasonable request and with permission of Ouranos Climate Centre.

Property data utilized during this study are available from the corresponding author upon reasonable request. The most up-to-date dataset can be acquired by sending a request to the Natural Resources and Energy Development department of New Brunswick.

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